Adaptive Task Selection in Automated Educational Software: A Comparative Study

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Abstract: In this study, we consider the challenge of adapting the difficulty level of the tasks suggested to a student using an educational software system. We investigate the effectiveness of different learning algorithms for the challenge of adapting the difficulty of the tasks to a student's level, and compare their efficiency by means of simulation with virtual students. Our results demonstrate that the methods based on Bayesian inference outperformed most of the other methods, while in dynamic improvement domains, the item response theory method reached the best results. Given the fact that correctly adapting the tasks to the individual learners' abilities can help them increase their improvement and satisfaction, our study can assist the designers of intelligent tutoring systems in selecting an appropriate adaptation method, given the needs and goals of the educational system, and given the characteristics of the learners.

Keywords: machine learning; educational technology; reinforcement learning, intelligent tutoring systems, adaptive task selection

1. Introduction

Given the advancement of technological capabilities, the field of distance learning has gained momentum. The ability of large groups of learners from all over the world to study together in borderless environments, is actually leading the concept of "learning" to a new, different, and fascinating place. On the other hand, it should also be noted that in such massive open online course (MOOC) environments, as well as in smaller learning environments, participants come from different groups, with heterogeneous prior knowledge and with heterogeneous abilities (Sein-Echaluce et al., 2016). Consequently, there is an important challenge of creating an environment adapted to student preferences and learning styles, and suggesting tasks adapted to the students' skills, abilities, tastes and needs.

Over the years, researchers have proposed various methods to address this issue (Beck, Woolf and Beal, 2000; Martin and Arroyo, 2004; L2TOR, 2020). However, customized education software and intelligent tutoring systems (ITSs), which can help stronger learners learn more quickly, while providing weaker learners assistance in overcoming

their difficulties, enabling all types of learners to progress at their own pace, have yet to reach their potential in this regard. This is due to the fact that most learning systems do not adapt the learning process to the student's characteristics. As a result, advances in this area of research can further improve the MOOCs performance, by facilitating adaptation of the topics taught to heterogeneous groups of learners.

The purpose of this chapter is to give an overview of different algorithmic methods for adjusting the level of educational software to the abilities of the student. In particular, we compare the performance of several learning algorithms in handling this important adaptation challenge. We consider the following adaptation schemes: Q-learning, virtual learning, Temporal Difference learning, Deviated Virtual Reinforcement Learning (DVRL), Bayesian inference, a Gittins-based method, and a method based on the Item Response Theory (IRT).

In order to enable the comparison of the adaption methods, we developed an artificial environment in which students were simulated, and they received tasks appropriate to their performance, using one of the adaptation methods evaluated. Using the above simulation, we compare the performance of the different methods, for different learning scenarios. Our study can assist the designers of intelligent tutoring systems in selecting an appropriate adaptation method, given the needs and goals of the ITS, the tutoring domain, expectations regarding students' skill improvements, and the pace of the improvements. Correctly adapting the ITS assignments to the individual learners' abilities can increase the learners' gains and satisfactions, and their motivation to successfully complete their learning sessions.

2. The Pedagogical Implications of Personalized Learning Systems

The idea of creating educational systems adapted to the different needs of students has fascinated many researchers from the early days of artificial intelligence (Carbonell, 1970; Smallwood, 1962) to the present. Personalized and customized education software that are able to detect the learner's context and adapt the learning material to this context can form a promising research field, and may transform the ways of learning and teaching (Hasanov, Laine and Chung, 2019). In particular, personalized educational systems can help stronger learners learn more efficiently, while providing

weaker learners assistance in overcoming their difficulties, enabling all types of learners to progress at their own pace.

When considering personal educational systems, several challenges exist. First, how to model the students and how to detect their abilities (Shawky and Badawi, 2018; Zhang et al, 2020), where special effort is placed on the task of handling the "cold-start problem": how to estimate the ability of new learners (Pliakosac et al, 2019). In addition, there is the challenge of adapting the form of learning to the student's abilities, strengths and weaknesses, and of adapting the learning progress in a way that is suitable to the student's profile (Huang et al, 2019) and adapted to the student's emotional stability (Alhathli, Masthoff and Beacham, 2020).

Another issue is the challenge of how to measure the hardness of the tasks presented to the learners (Hutzler et al, 2014); Benedetto et al, 2020;

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Xue, K., Yaneva, V., Runyon, C., and Baldwin, P. (2020).

, Pankiewicz, M. and Marcin Bators, M. (2019).

]. An additional important challenge is how to create a personal environment and supportive behaviours tailored to the student's needs and behaviours [Gao, A. Y., Barendregt, W. and Castellano, G. (2017).,

Gordon, G., Spaulding, S., Westlund, J. K., Lee, J. J., Plummer, L., Martinez, M., Das, M. and Breazeal, C. (2016), Yang, T. C., Hwang, G. J. and Yang, S. J. H. (2013). Development of an adaptive learning system with multiple perspectives based on students' learning styles and cognitive styles. *Educational Technology & Society*, 16 (4), pp. 185–200.

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Personalized learning can be effective in various areas of computer-assisted learning such as learning new material by means of on-line courses [Conati, C., Gertner, A. S., VanLehn, K. and Druzdzel, M. J. (1997).,

Conati, C., Gertner, A. and VanLehn, K. (2002).,

Sein-Echaluce, M. L., Fidalgo-Blanco, Á., García-Peñalvo, F. J. and Conde-González, M. Á. (2016).
Kaabi, K., Essalmi, F., Jemni, M. and Qaffas, A. A. (2020).,
Ewais, A. and Abu Samara, D. (2020).,
Kakish, K. and Pollacia, L. (2018)]:
The adapted learner can choose which topic to present to the student, how to present it, and how to decide that the topic has already been studied properly and can proceed with new material. Here, the student's model basically will be his level of knowledge in each of the subjects and can also include the most appropriate mode of absorption for him.
When considering training software, such as software to improve reading and reading comprehension skills [Shamir, H., Feehan, K., and Yoder, E. (2017)., Azoulay, R., David, E., Avigal, M. and Hutzler, D. (2020).], or systems to rehabilitate cognitive abilities [Kizony, R., Katz, N. and Weiss, P. L. T. (2003). , Rossol, N., Cheng, I., Bischof, W. F. and Basu, A. (2011), , Vaughan, N., Gabrys, B. and Dubey, V. N. (2016).], the system can decide at each stage which exercise will suit the students' ability. This is done to train the students in such a way that on the one hand will challenge them and improve their abilities but on the other will not discourage them. In both scenarios, the adaptive system can adapt the learning units and tasks and their presentation according to the personal profile of each learner.
The adaptivity of an educational system can apply to several aspects, including: choosing the learning topics and deciding when to move from topic to topic; choosing the best learning strategy for each student; adjusting the difficulty or the type of challenges and material presented according to the level of the student; providing clues tailored to the student's level, in a way that will help him meet the challenges he faces; presenting feedback adapted to the student's level and style of understanding; adapting the display and structure of the system to the student's way of thinking and comprehension, and in light of the student's preferences.
Software adapted to the student's profile, preferences and abilities can bestow the student a better feeling and can cause greater engagement in the course as well as improved achievements, compared to a non-adaptive course [Kulik, J. A., and Fletcher, J. D. (2016).].
Therefore, it is crucial to compare different adaption methods that can be used, in order to offer appropriate recommendations for different educational goals and situations. In

this study, we concentrate on the aspect of adapting the challenges provided by the system to the needs and abilities of the individual learner.

We performed this type of comparison by simulating different methods of adapting the level of challenges to the level of the student, given the belief that adapting the challenges in educational systems can be useful and helpful in all educational cases and for different sizes of learning platforms.

3. Overview of the Different Adaptation Schemes

The core of each personalized education system is the mechanism that applies relevant tasks and challenges to each learner. In this section, we survey several studies concerning machine learning methods to adapt the software tasks and behaviour according to the student's abilities.

In general, there are two main paradigms to consider in the adaption goal: The first paradigm contains methods based on reinforcement learning, and the second paradigm contains methods based on Bayesian inference. In the rest of this section we will survey several methods for these two paradigms.

3.1 Reinforcement Learning Methods

Several adaption algorithms widely used in ITS are based on Reinforcement Learning (RL) algorithms [Sutton, R. S., and Barto, A. G. (1998).,3], which are a machine learning method developed for sequential decision-making under uncertainty with delayed rewards. In RL, the sequential decision making problem is defined as a Markov Decision Process (MDP). Formally, a finite-horizon MDP is defined as a five tuple (S, A, T, R, H), where S is a set of states, A is a set of actions, T is a transition function where $T(s' \mid s, a)$ denotes the probability of transitioning from state s to state s' after taking action a, $R(s, a)$ is a reward function, which specifies the reward (or the probability distribution over rewards) when action a is taken in state s, and H is the number of time steps where actions are taken. In each time step, the RL agent chooses an action $a \in A$ from the set of available actions. The environment moves to a new state $s \in S$, and the reward associated with the transition is determined. The goal of the agent is to collect as high a reward as possible. The main dilemma of the agent is the "exploration-exploration dilemma," in which the agent must decide whether to choose

the currently best known action or try taking other actions in order to gather more information that might improve the agent's decision-making capabilities.

In reinforcement learning (RL), the goal is for an agent to learn a policy π—a mapping from states to actions or probability distributions over actions—that incurs a high reward [Sutton, R. S., and Barto, A. G. (1998).].

The reinforcement learning methods are, in general, heuristic methods that learn the best performed action to be used in each situation (state), and their advantage is their low complexity.

A recent review of the usage of RL for adaptive instructions to students is presented by Doroudi et al. [Doroudi, S., Aleven, V. and Brunskill, E. (2019).], who claim that the most successful adaptive instructions systems were those designed based on theories from cognitive psychology and learning science.

However, given that the theories are limited, they also find that it has been useful to complement this approach with running more robust offline analyses that do not rely heavily on the assumptions of the theories. We proceed by providing the details of several reinforcement learning methods that can be used for task adaptation.

3.1.1. ***Q-Learning***

The Q-learning algorithm [33,34,3,35] is an RL algorithm, in which, the Q-learning algorithm saves a value *Q* for each pair <*s*,*a*>, where *s* is the current state, and *a* is the particular action taken by the agent given the current state. Given the Q values, with a probability of ε, the algorithm explores and randomly chooses an action, and with a probability of 1-ε, the algorithm exploits and chooses the action with the highest Q value. In our framework, action *a* denotes the question level.

As soon as action a is taken and the reward is observed, the Q value of action *a* is calculated using formula (3):

$$Q(a) \leftarrow Q(a) + \alpha(r + \gamma \cdot max_{a'} Q\left(a'\right) - Q(a)\ ) \tag{3}$$

where α defines the speed of convergence of the Q values, α=0 results in the agent not learning anything, while α=1 results in the agent considering only the most recent information, *r* is the reward value, and γ defines the discount factor, where *γ=0* will cause the agent to consider only current rewards.

In our study, only one state exists, and the actions are the possible question levels. Each question level is considered a possible action and is associated with a certain Q value. Once the student provides an answer, the relevant Q value is updated according to the reward, which indicates the success or failure in answering the question. In our simulation, the parameters used in the Q-learning methods (TD learning, VL, and DVRL) were initialized as follows: ε=0.1, λ=0.95, α=0.5, and the Q array values were initialized as one.

3.1.2 Temporal Difference Learning

The temporal difference (TD) learning algorithm [36] is a variation of the RL algorithm. It is different than the Q-learning algorithm in terms of the way the future reward is calculated. For example, in the SARSA algorithm, which is a TD variation, the updating rule uses the following formula:

$$Q(a) \leftarrow Q(a) + \alpha[r + \gamma \cdot Q(a') - Q(a)] \quad (4)$$

where in our study, a and a' are the old and new question levels, and $Q(a')$ is the value function of choosing question level a', using the policy applied by the TD algorithm to choose an action. At each time point, the next question's level is chosen by the TD algorithm's policy, which again choses a random question level with a probability of ε, and the level with the highest Q value is chosen with a probability of 1-ε. Then, the immediate reward r is calculated, using evaluation function 1 or 2. Finally, the Q values of all possible decisions (levels) are updated according to formula (4).

3.1.3 Virtual Learning

Virtual learning (VL) [37] is also similar to Q-learning, but in this case, instead of learning only on the basis of actions and rewards actually experienced, the algorithm can also learn by reasoning from the outcome of one action in order to predict the expected outcome of other actions.

In our study, once a student succeeds in answering a question, the Q value of the current level is increased, as are those of the lower levels. Similarly, if a student fails to answer a question, the Q value of the level of the current question is reduced, as are those of the higher levels.

The VL variation called DVRL (Deviated Virtual Reinforcement Learning) [38,39] is similar to VL, but once a reward is received for the student's answer, the updating phase of the Q values relates not only to the given level and the levels in the appropriate direction but also to the level of the neighbouring questions. Specifically, once a student answers a question correctly, the Q-learning of the closest higher level also increases, and when a student fails to answer a question correctly, the Q-learning of the closest lower level also decreases.

3.2 Bayesian Inference Based Methods

Several studies have proposed using Bayesian-based methods in ITSs. All of them are based on certain beliefs about the student's abilities, where over time these beliefs are updated given the student's performance, using the Bayesian rule [40, 23], or using some other similar technique that is based on updating the beliefs about students' abilities. In this section we survey several systems that were implemented based on Bayesian techniques.

3.2.1 Bayesian Inference Algorithm

According to the Bayesian inference method, at each particular time, the actual level of the student is unknown to the system, but it is known to have a normal distribution with a mean and a variance. Furthermore, the ability of the students to successfully answer a particular question depends on their actual level at that particular time. We assume that that student's abilities are defined by a mean and a variance of the distribution of the level of each student. Moreover, we assume that the student's level can be represented by an integer between zero and five, and the standard deviation of the student’s level can be an integer between zero and five. Given a particular student, the algorithm considers different possible parameters of the distribution of the student's level. Initially, the algorithm associates a constant probability with each set of parameters (μ and σ). In each step, the best alternative task is chosen given the current beliefs about the student's distribution, and after the student's response is observed, these beliefs are updated. Additional details about the Bayesian Inference algorithm are presented in Appendix A. This variation of the Bayesian inference method was successfully used for training high school students in reading comprehension skills [Azoulay, R., David, E., Avigal, M. and Hutzler, D. (2020).].

3.2.2 Bayesian knowledge tracing

Bayesian knowledge tracing (BKT) is a technique widely used in ITSs to model a student's skills in various subjects and continuously update the probability that the students have mastered a skill as their work progresses [41]. There are four types of model parameters used in Bayesian knowledge: the initial probability of knowing the skill a priori, the probability of the student's knowledge of a skill transitioning from the unknown to the known state after an opportunity to apply it, the probability to make a mistake when applying a known skill (slip), and the probability of correctly applying an unmastered skill (guessing). Bayesian inference is used in order to update the above parameters given the ITS observation, and the ITS's decisions are made based on those values. The model's parameters can be learned generally, assuming the same parameters exist for all learners, or they can be individualized for each learner.

Against this background, we emphasize the differences between BKT and the Bayesian inference we used, which include three main ones: (a) In the BKT model, each learner is characterized by a set of variables, where each variable defines the knowledge of one subject or topic. In our study, we consider the training of a certain skill, and the learner is characterized by a distribution (assuming a normal distribution) of the student's abilities regarding this skill. (b) In most of the BKT studies, the variables are binary and specified for each topic, whether or not the topic was already learned by the student. In contrast, in our study, the variables inferred using the Bayesian inference rule are the mean and variance of the student's skill distribution. (c) The BKT model considers the influence of each action on the student's knowledge on each topic, while in our study, if the learner's skills change during the ITS, it will be inferred over time based on the student's performance.

3.2.3 Bayesian networks

Bayesian networks are a class of graphical models defined over a set of random variables, each describing some quantity of interest. The structure of the network encodes the independence relationships between those variables, with graphical separation implying conditional independence in their probabilities.

Several educational systems use Bayesian networks to identify the specific learner's characteristics. Kuila et al. [42] built an ITS that uses a Bayesian network to choose

both the difficulty level of questions during a problem solving session and the appropriate responses to the student during the problem solving process. The authors emphasized that the performance of the system greatly depends on the Bayesian network topology, which is automatically updated during the learning process. Conati et al. [Conati, C., Gertner, A. S., VanLehn, K. and Druzdzel, M. J. (1997)., Conati, C., Gertner, A. and VanLehn, K. (2002).] used a Bayesian network for long-term knowledge assessment, plan recognition, and prediction of students' actions during problem solving, for Andes, which is an ITS for Newtonian physics. The goal of the Bayesian network is to provide hints and guidance to the student during the learning process.

Ramírez-Noriega et al. [43] developed an evaluation module for use in an ITS based on Bayesian networks. They present a question-based evaluation design and provide algorithms for moving the student through the levels of questions according to the knowledge of the course concepts. The authors tested their evaluation method on an undergraduate Algorithm and Computer Logic course held at the University of Sinaloa, Mexico. They compared the effectiveness of the ITS-based evaluation module with a traditional written exam, and their results show that the ITS test was more effective.

3.2.4 Multi-Armed Bandit based Methods

Another statistical approach used for deciding the optimal sequence of activities considers the task difficulty selection challenge as a multi-armed bandit (MAB) problem and uses an appropriate MAB to solve it [44,45 ,46,47].

For the original MAB problem, Gittins [48] developed a tractable method for solving the multi-armed bandit problem and calculated indices specifying the attractiveness of each arm, including future rewards from exploring unknown arms. In our study, Gittins indices are used to compare the success rate of different levels. In addition, the Gittins indices method used in this study takes the average and standard deviation of the past results for each level into consideration, in the manner suggested in [49]], where the Gittins indices algorithm is applied to multiple arms, and each arm is normally distributed with any value of mean μ and standard deviation σ. Focusing on our problem, Algorithm 1 is used to choose a question level, given GittinsIndex[n], which is the array of Gittins indices for each number n of past trials of an arm, and given the following arrays saved for each student: n[level] - the number of past trials of each level

by the student; sum[level] - the sum of rewards for this level; and std[level] - the standard deviation of rewards for this level.

Algorithm 1. Gittins indices-based algorithm

IF (a level with *n[level]*≤2 exists) then

choose a level *level'* such that *n[level']* ≤2

ELSE choose a level *level'* which maximizes *value(level')*

Where $value(level') = sum[level'] + std[level'] * GittinsIndex[n[level']]$

In other words, each level of questions is considered an arm, and at each stage, Gittins indices are calculated for each possible question level, and the level with the highest Gittins index is chosen.

3.2.5 Item Response Theory and Personalized Testing Methods

Item response theory (IRT) consists of statistical models that relate item responses to the abilities that the items measure [50,51]. In the most basic model, the one-parameter Rasch model, the probability of a correct answer is a logistic function of the difference between the ability of the subject and the difficulty of the question. In the two-parameter logistic model, the difference is weighted by an item discrimination parameter, which has a high value when the probability of a correct response increases more rapidly as the ability (latent trait) increases. In addition, the three-parameter model uses the extent to which candidates can guess the correct answer. Item response models can be used for equating tests, to detect and study differential item functioning (bias), and to develop computerized personalized testing.

Wauters et al. [52] proposed monitoring a learner's proficiency level in the ITS by using the Elo rating system, originally used for chess and successfully implemented by [53,54,55] in the educational field. They successfully use a logistic function to adapt the weights of the Elo rating system, and their results on real datasets indicate that the Elo algorithm with a logistic weight function better approximates the parameter estimates obtained with the item response theory than the Elo algorithm with a fixed weight. To summarize, methods used for computer aided tests based on item response theory can be useful for the task of adaptive level selection in ITSs.

Papoušek et al. [54], use the CAT model to improve students' skills rather than to test them. The Rasch model (one-parameter logistic model) uses a cumulative distribution function for a correct answer, given a skill parameter θ of the student and the item difficulty b. According to this model, the probability of an answer to be correct is denoted by

$$\text{P(correct} \mid \theta), \text{ is } \mathrm{P}(correct \mid \theta) = 1/\left(1 + e^{-(\theta - b)}\right).$$

In addition, updating P, given a correct or an incorrect answer, is done with the Elo rating system, by using the following updating formula after each answer:

$$\theta = \theta + U(n) \cdot (correct - P(correct \mid \theta))$$

where correct is zero if the last answer was wrong and one if the last answer was right, and *U(n)* denotes the uncertainty function, which determines the sensitivity of the estimate of the last attempt. *U(n)* is defined as $U(n)= \alpha/(1+\beta n)$ where n is the number of previous ITS rounds for the current student, and α,β are meta-parameters that were optimally fixed as α=1,β=0.06 in the empirical results of Papoušek et al. [54].

In our study, we suggest using the model suggested by [53], and θ initialized as one for a student and updated after each round.

The IRT was originally defined for one ability parameter, but the recent study of Park et al. [56] extends the standard ERS that updates a single ability parameter based on the Rasch model, and allows simultaneous updates of multiple ability parameters.

Generally, IRT is used for diagnostic purposes. Piech et al. [57] suggested a formulation called Deep Knowledge Tracing (DKT) in which flexible recurrent neural networks are applied to the task of knowledge tracing. Using deep learning techniques enhances traditional IRT for cognitive diagnosis by exploiting sematic representation from question texts with deep learning.

Based on the DKT method, Yeung [58] developed Deep-IRT, which is a synthesis of IRT and knowledge tracing, aimed to make the deep learning based knowledge tracing model explainable. Specifically, a Dynamic Key-Value Memory Network (DKVMN) [59] was used to process the student's learning trajectory and estimate the student ability level and the item difficulty level over time. Then, the IRT model was used to estimate the probability that a student will answer an item correctly using the estimated student ability and the item difficulty. Experiments showed that the Deep-IRT model retains the performance of the DKVMN model, while it provides a direct psychological interpretation of both students and items.

3.3. Summary of the Comparison of Adaptive Methods

Based on the above detailed description of adaptive methods, that can be used to achieve a customized educational system, it is noteworthy that the different adaptive methods can vary in their complexity but also in their adaptive behaviour: There are adaptive methods that tend to present minor challenges, following an under-estimation of the student level, whereas there are methods that will present more accurate challenges, which may also discourage the student. Another difference is in the level of resources required to run the method, with some methods being very simple to implement, and others consuming more memory and time. In addition, an important issue is the ability of the method to be suitable for extension to a situation where the student is not modelled by one feature, such as his level of understanding, but by a vector of abilities, or a vector of his knowledge on various topics. In this case, there are methods that are readily suitable, or have already been adapted for use in personal learning systems, while the adaptation of other methods may be complex. In addition, there are methods that are more appropriate to situations such as games, where different states exist, depending on students responses and possibly on additional factors.
Table 1 summarizes the main properties, strengths and weaknesses of each of the methods described above.

Algorithm	***PROC***	***CONS***	***SUGGESTED DOMAINS***
Q-learning **TD** **VL, DVRL**	Efficient in time and space, Easy to implement, appropriate for multiple sates.	No accurate modeling of student abilities. Task tends to be easier than with other methods. Hard to extend to multiple attributes of students.	Learning games, where different states exist according to learner actions; Situations where the weak learners may leave the system if the task is too challening.
Bayesian inference	Relatively exact estimation of student ability.	Relatively high time and space complexity.	

Bayesian network	Defined for situations where student has multiple attributes.	Building the Bayesian network topology is an NP-hard problem: time and space complexity may be challenging.	Educational system for studying new material, where absorption of the material depends on the different characteristics of the learner, and a complex student model is required.
Gittins based method	Low complexity, Appropriate for choosing between alternatives, relatively accurate results for dynamic changes in students' levels and abilities.	Is less appropriate when different states exist.	Steep learning curve, Relatively simple student model, relatively simple educational system.
IRT, Elo updating rule	Low complexity, high accuracy, easy for implementation, was recently extended for domains with several dimensions of student's abilities	Does not consider different states of the game/system.	Steep learning curve, Deep IRT can be used to extract student's hidden skills and properties.
Deep knowledge tracing, Deep IRT	It is able to build complex student models given the student actions, without the need to define it in advance.May be appropriate for the multiple state model.	Needs a large set of examples in order to be trained, highly complex in the training phase, high space complexity	Suitable for an online education environment, where large amounts of training data exist

As depicted in Table 1, each method has its own advantages and drawbacks, and the best method should be chosen given the exact situation. In particular, for online education environments, where large datasets exist, deep learning methods may be applicable [60], while in small classrooms, simpler methods can be more appropriate. In addition, the type of students and aim of the system (training of skills or learning new

topics) may have an effect on the student learning curve and, as a result, on the relation between the performances of the different methods. Third, the structure of the educational system, whether different states exist, such as in educational games, or only one state exists that is observed, may also effect the decision regarding the best method to use.

In the rest of this chapter, we compare situations where one state exists, for different adaptive methods, for one dimensional skill of the student. Nonetheless, given the different features and characteristics of the methods, and given the fact that there are more complex methods, it is worth proposing the construction of a modular educational system, in which the module responsible for adjusting the level of questions can be replaced, if necessary.

4. Methodology and Research Approach

In the rest of this chapter, we concentrate on the task selection problem in an educational system, where the goal of the adaptive algorithm is to determine the level of the next question, and given this, the ITS, in turn, should choose an appropriate task in that level to be presented to the student. We proceed by specifying the methodology used in our simulation in order to be able to compare the different adaptive methods. When working with the ITS, each student obtains a set of N questions. The task selection process is based on three steps: (1) initialization of the student's model; (2) the process whereby the system chooses the next appropriate question; (3) examination of the student's answer and saving the information for future steps.

The aim of the ITS is to utilize and sharpen the learners' skills by presenting questions that are appropriate for the students – questions that the students can successfully answer, while being at the highest level that allows them to do so. Thus, when considering the task selection problem as an optimization problem, the criteria we would like to optimize should be reflected by an appropriate evaluation function, and the evaluation function should also be used to assess the algorithms' performance.

4.1 The Evaluation Functions Used in our Study

In this study, we utilize the following two evaluation functions:

Evaluation function 1 sums the level of the questions that were successfully answered by the student, where the higher the level of questions answered successfully, the higher the total evaluation. In other words, the utility of a successful answer is equal to the question level, and the utility of a failure (i.e., an incorrect answer) is zero. Evaluation function 1 is specified in Equation (1).

$$evaluation_1 = \sum_{question_i=1}^{N} f_1(question_i) \quad (1)$$

where

$$f_1(question_i) = \begin{cases} Level(question_i) & question_i \text{ was answered correctly} \\ 0 & question_i \text{ was answered incorrectly} \end{cases}$$

Evaluation function 2 also considers the level of the questions, but in contrast to the above measurement, this function adds a constant reward $C_{success}$ for each success and a constant penalty $-C_{failure}$ for each failure; incorrect answers are reflected in a decrease in the total evaluation. Evaluation function 2 is specified in Equation (2).

$$evaluation_2 = \sum_{question_i=1}^{N} f_2(question_i) \quad (2)$$

where

$$f_2(question_i) = \begin{cases} Level(question_i) + C_{success} & question_i \text{ was answered correctly} \\ -C_{failure} & question_i \text{ was answered incorrectly} \end{cases}$$

Both evaluation functions are applied for the set of questions presented to the student, and both evaluation functions are utilized in our simulation, in order to calculate the reward based on the student’s answer and in order to evaluate future rewards. The motivation behind these two functions is as follows. Evaluation function 1 considers the advances in the student's knowledge, by considering the sum of levels of the questions that were successfully answered. However, evaluation function 2 also considers the preferences of the student who would prefer to answer the questions correctly, since each success increases the evaluation function, and each failure decreases it by a constant value, regardless of the level of question that was asked.

Note that evaluation function 2 may, in some cases, yield a negative value, which means that the total gains of the student from the system could be negative, for students with large failures. This means that the situation of such students was better prior to working with the system, since after working with the ITS and failing to correctly answer the questions, the students' confidence will likely be lower, while knowledge and skills will not have improved. Note that when *Csuccess=Cfailure=0*, the resulting function has the

structure of evaluation function 1. Evaluation function 2 places more emphasis on the student's success and failure in answering questions, by adding a constant value or deducting a constant value from the evaluation function for each success or failure, respectively.

We note that in our study, the students are unaware of the evaluation measurement calculated by evaluation function 1 or 2, and they are only aware of the grade of each of them based on the tasks each of them succeeded in solving.

4.2 A Description of the Simulated Environment

In order to compare the performance of the algorithms mentioned above, we developed an artificial environment with simulated students for the evaluation phase.

In the simulation, each student has a distribution of the skill/knowledge level at each step. The distribution of the student's level is a Gaussian function, with a given mean and standard deviation. The process of generating the behaviour of each student is expressed as follows:

1. When a new student is generated, the distribution parameters (mean μ and standard deviation σ of the student's level) are drawn randomly from the range of 0..5.
2. Then, a vector of *H* thresholds is created (where *H* can be 10,20 or 100):
    - Where each threshold is randomly drawn from the normal distribution, given the mean μ and the standard deviation σ of the new student.
    - If an improvement rate $\delta>0$ exists, then the mean parameter of the student becomes $\mu'= \mu*(1+\delta)$.
3. For each adaptive method, the same vector of thresholds is used.
    - At each step, the adaptive method suggests the difficulty of the next task, and the difficulty is compared to the next threshold saved in the predefined threshold vector.
    - If the task difficulty is higher than the predefined current threshold, then a failure occurs.
    - Otherwise, a success is reached.

- Given the result of the current step, the adaptive strategy updates its beliefs or its internal knowledge, in order to suggest the task difficulty of the next step.

The above process (steps 1-3) is run N times, where in each run, a new random student is created (with the *H* pre-generated thresholds), and each adaptive method is run for the same list of *H* pre-generated thresholds. In each run, the statistics (success rate, value of evaluation functions) for each student are collected for each adaptive method.

Finally, after completing N runs, the average and standard deviation of the results over the N students is calculated for each method.

The goal of the ITS is to choose the appropriate questions for each student, so that the number of questions the student answers correctly increases and the questions' level of difficulty remains challenging. In order to be able to evaluate the combination of these two maximization criteria, we proposed evaluation function 1 and 2, as defined in Section 4.1. These evaluation functions reflect the fact that the higher the level of the question and the more correct answers, the higher the utility. Thus, it takes both aspects into consideration. But, when using evaluation function 1, an incorrect answer provides zero reward. While evaluation function 2 is used to check the negative emotional effect in case of failure. Once the student finishes using the ITS software, the total utility value of the student is updated and saved.
Given the procedure described above, we are able to evaluate each of the adaptive methods on the same set of inputs (sets of current levels/thresholds, as described above).

5. Simulation Results

We ran simulations for an ITS that aimed to present 10, 20, and 100 questions to the student. For each number of questions presented, and for each randomly created student, we ran each of the proposed adaptation algorithms for 10,000 runs, using evaluation function (1), where the utility of a successful answer is equal to the question level, and the utility of a failure (i.e., an incorrect answer) is zero. The simulation results are presented in Table 2. In each cell, the upper-left value represents the average utility achieved by the given algorithm for the particular number of questions; the upper-right value (in parentheses) represents the standard deviation of that average; and the bottom

value represents the ratio between the average utility obtained by the algorithm in the simulated environment and the average utility obtained by the ITS based on the student's known level of distribution for each time slot.

Table 2. The average utility, standard deviation, and average utility ratio obtained by the adaptive algorithms

Algorithm	10 questions	20 questions	100 questions
Q-learning	10.779 (6.39) 74%	22.19 (12.16) 75%	112.174 (57.04) 77%
Virtual Learning	10.59 (6.24) 73%	21.544 (11.86) 73%	104.223 (54.5) 71%
TD	10.811 (6.41) 74%	22.219 (12.21) 75%	114.273 (59.09) 78%
DVRL	10.419 (5.98) 72%	20.958 (11.35) 71%	103.448 (54.05) 71%
Gittins algorithm	10.09 (6.14) 69%	22.218 (13.36) 75%	131.985 (74.87) 90%
Rasch model, Elo updating rule	11.8 (6.93) 81%	24.7132 (14.34) 84%	127.37 (71.84) 87%
Bayesian inference	**12.99 (8.27)** **89%**	**26.632 (15.57)** **90%**	**135.663 (74.04)** **93%**

As shown in Table 2, the Bayesian inference algorithm obtained the highest average utility compared to the other algorithms, with 89-93% of the utility gained by a theoretical optimal algorithm that has complete information about the real distribution of each student. The second and third highest average utilities were achieved by the Rasch+Elo method, based on the IRT, and by the Gittins-based methods. These results can be explained by the fact that the Bayesian inference algorithm considers several possible models (distributions) of each student and updates the probability of each of them and thus obtains more accurate results than the other methods.

5.1 Top Performing Algorithms

We now look more closely at the three top performing adaptive algorithms to gain some insights. When comparing the Rasch+Elo method to the Bayesian inference method, we

see that these two methods use a similar probability function: the Rasch cumulative distribution function used in the Rasch+Elo algorithm can be considered an approximation of the cumulative distribution function of the standard normal distribution. In addition, in the Bayesian Inference method, a probability is saved and updated for each possible model of the student, while in the Rasch+Elo method, the updating rule saves one parameter for each student, which represents the student's skills, which parameter is updated over time, and provides the new results (successes and/or failures) of the ITS. When comparing the Gittins indices algorithm and the Bayesian inference-based algorithm, we observe that both methods assume normal distribution of the alternatives. The advantage of the Gittins indices algorithm is that it takes future rewards into account as a result of the exploration of new arms, considering an infinite horizon scope, given a constant discount ratio, while the Bayesian inference algorithm maximizes the expected utility of the next time step. The advantage of the Bayesian inference algorithm is that it considers the correlation between different levels, whereas the Gittins-based algorithm assumes independent arms. In fact, the Bayesian inference-based algorithm outperformed both the Rasch+Elo and the Gittins-based methods. However, in situations where the student's skills improve over time, as demonstrated in Figure 5.1 below, the Rasch+Elo method outperformed all of the other methods, due to its flexibility with regard to changes over time, followed by the Gittins method, and the Bayesian Inference method which came in third place. Another advantage of these methods over the Bayesian inference algorithm is the fact that they require fewer resources.

5.2 Comparison of Algorithms Over Time

Next, we compare the average utility and the rate of success, which is presented in Figures 5.1 and 5.2. Figures 5.1 and 5.2 depict the students' progress over time and the algorithms' ability to take it into account. As illustrated in Figure 5.1, the DVRL and Q-learning methods achieved the highest results in terms of the students' ability to answer the questions, but since they achieved a relatively low average utility, we can infer that these methods tend to present easy questions relative to the optimal level of questions that should be provided to the students.

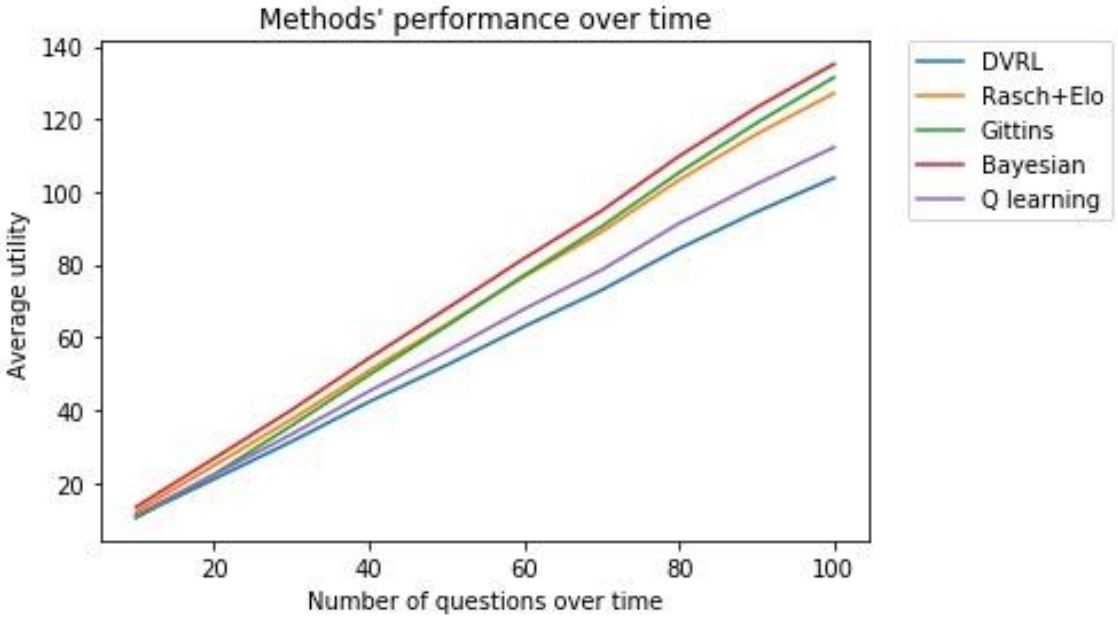

Figure 5.1: *The average utility as a function of the number of questions.*

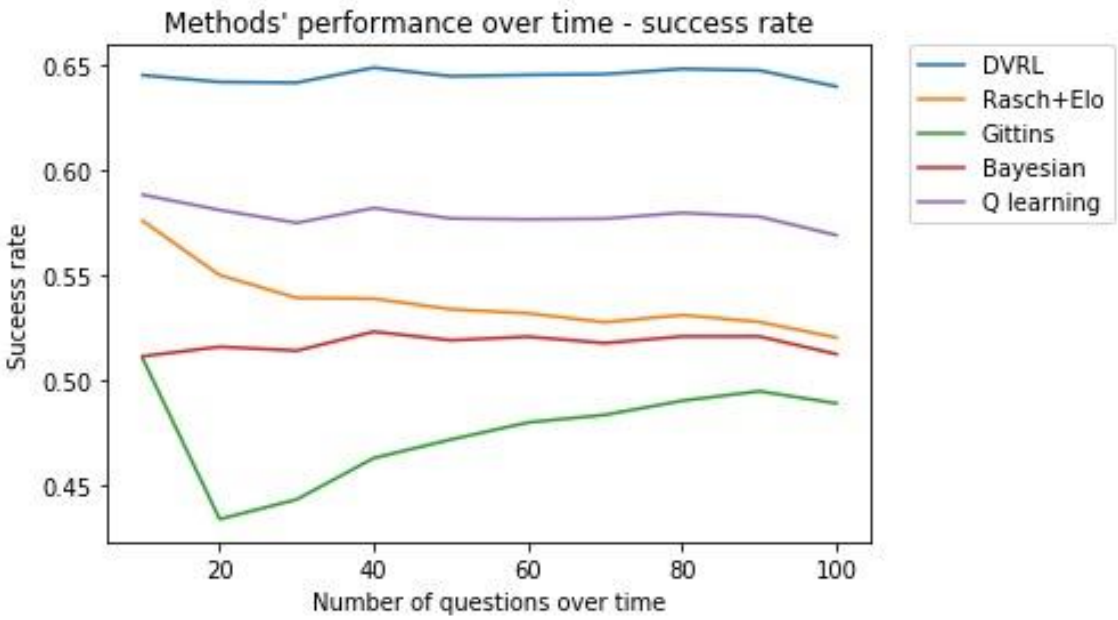

Figure 5.2: *The rate of successfully answered questions as a function of the number of questions.*

5.3 Comparison of Algorithms Using Different Evaluation Functions

In addition, we checked whether the rate of success of the Bayesian inference algorithm can be increased by assigning a negative utility to a question answered incorrectly and a positive utility to a question answered correctly, regardless of its level, as provided in

evaluation function (2) (defined in Section 2.1). According to evaluation function (2), for each question answered correctly, the student evaluation function increases by the value of the question level plus a constant $C_{success}$ (reward), and for any question answered incorrectly, the evaluation function decreases by the value $C_{failure}$ (penalty). In our simulation, $C_{success}$ was equal to $C_{failure}$, but it should be noted that the constant value can be different if it better reflects the ITS designer's preferences. Consequently, we compare the proposed algorithms when evaluation function (2) is used (instead of evaluation function (1) illustrated in Figure 5.1 and 5.2), and the results are presented in Figure 5.3 and 5.4.

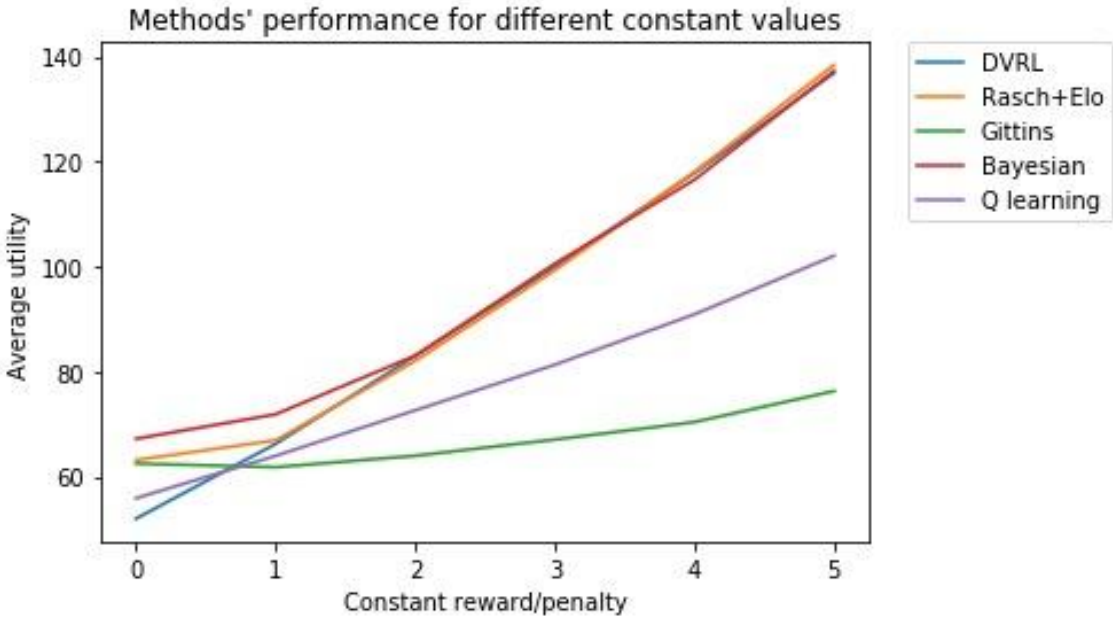

Figure 5.3: *The algorithms' average utility as a function of the constants* $C_{success}=C_{failure}=0,..,5$ *used in evaluation function (2).*

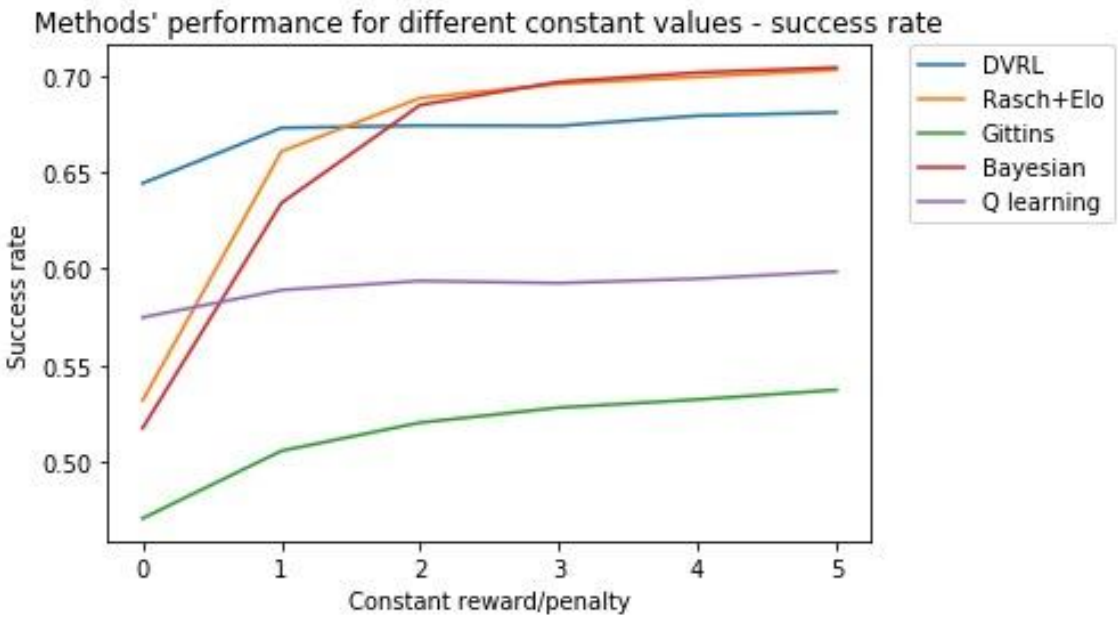

Figure 5.4: *The rate of questions correctly answered as a function of the constants* $C_{success}$ *and* $C_{failure}$.

As depicted in Figure 5.3 and 5.4, for an evaluation function with a constant reward for success, and a penalty for failure $C_{failure} \leq 3$, the results of the Bayesian inference algorithm and those of the Rasch+Elo method provided the highest average utility, where for higher values of constant rewards and penalties, the DVRL method (a variation of the Q learning method) reached similar results. In addition, the Bayesian inference algorithm and the Rasch+Elo method reached the highest success rate, for each penalty level from 2, while for lower penalty, the highest winning probability is reached by the DVRL method. The explanation for these results lies in the fact that the DVRL method tends (by average) to choose questions of a lower difficulty level, for different levels of penalties, thus when a penalty for failure is low, this method reached the highest success rate. But as the penalty, as well as the gain for correctly answering the task, increases, the Rash+Elo and the Bayesian method adapt themselves to choose tasks that are easier than those chosen by the DVRL method, and as a result, their success rate is the highest, as well as their average utility (the average value of the evaluation function) when using these methods.

To conclude, the results shown in Figures 5.3 and 5.4 demonstrate that the choice of the evaluation function has great impact on the comparison of the algorithms, and further research should be conducted to verify which evaluation function optimally represents

the effectiveness of an ITS, taking into account the learners' preferences, in order to provide the best individualized tutoring experience possible.

5.4 Algorithm Comparison for Dynamically Changed Level of Students

Another interesting issue relates to the algorithms' performance for students who dynamically improve their skills during their training on the ITS. In order to evaluate the performance of the proposed algorithms, in this case, we assumed that a mean student distribution level for each student would increase at a rate of 1+δ after each time step of the simulation (i.e., after each question). Figure 5.5 demonstrates that for runs of 50 questions, where the mean level of the student improved in each step by a rate of δ=0.01 or more, the Rasch+Elo method outperformed the Bayesian inference method, and when the improvement rate was δ=0.012 or more, the Gittins-based algorithms came in second place, while the Bayesian inference method came in third place.

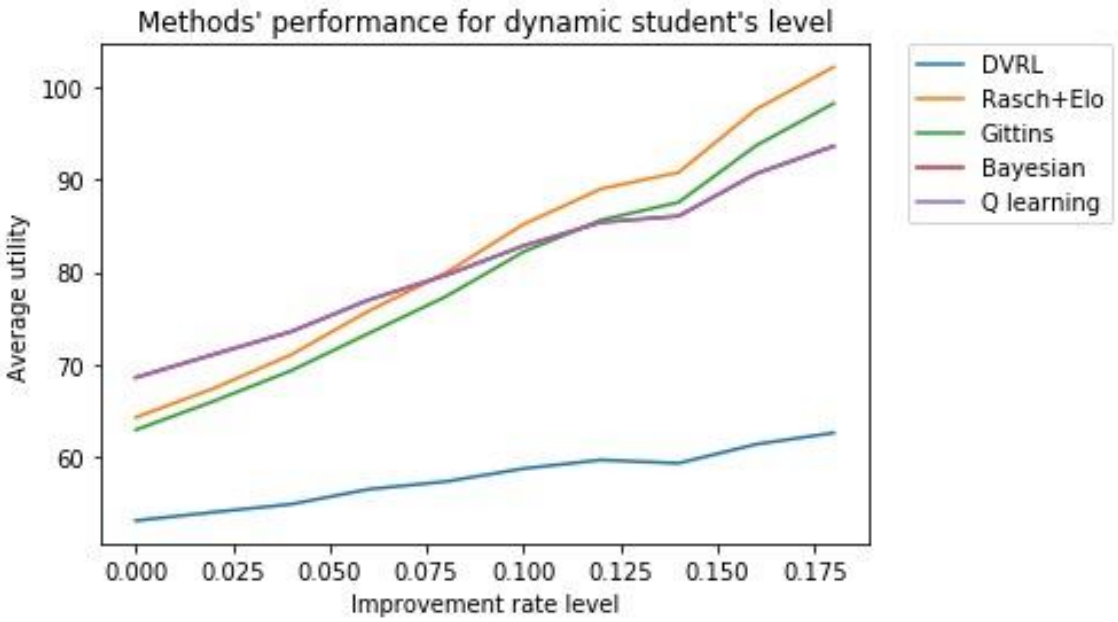

Figure 5.5*: The algorithms' average utility for different improvement rates in the student's level over time.*

The reason for this lies in the fact that the Rasch+Elo method gives significant weight to the last result (success or failure) and therefore is more affected by changes in the user's level. In addition, the Gittins-based algorithm may return to non-beneficial arms (options) when the arms' usage becomes relatively low (since the Gittins-based method

depends on the arms' usage). Thus, the Gittins-based algorithm does not ignore the improvement of the student, and gives the ITS the ability to present harder questions as the student's level improves. Note, however, that an improvement rate of δ=0.01 means that after 50 questions, the students will by average improve their level by 64% ($1.01^{50} = 1.64$), and an improvement rate of δ=0.012 means that after 50 questions, the student's level will be improved by 81.5% ($1.012^{50} = 1.815$), but in real environments, such an improvement is not realistic after such a short time period.

To summarize, we can see that in environments with relatively gradual changes in the student's level, the Bayesian inference method obtains the highest utility in most of our experiments. With dynamically changing levels, the Rasch+Elo method based on the IRT is the best method, and the Gittins-based method follows as the second best method, due to their flexibility with regard to changes in the students' level, and the Bayesian inference algorithm comes in third place. The advantage of the Bayesian inference method is the fact that it considers different possible distributions for each student, but the advantage of the Rasch+Elo and Gittins methods is their sensitivity to changes and the fact that they can be easily implemented and require fewer resources than the Bayesian inference method.

6. Discussion

In this chapter, we describe several algorithmic methods that can be applied to adapt the task hardness to the learner's current abilities. In order to compare the different methods, we created simulations of students to solve tasks in an educational system, and we compared the performance of different methods in terms of two types of evaluation functions which take into consideration the advantage of high level tasks, where the second evaluation function also considers the "cost" of failing to correctly answer the task, which cause a reduction in the student's motivation. When using one of these two evaluation functions, we compared the performance of the different methods by means of simulation, and checked their ability to adapt to the learners' abilities.

The results show that the Bayesian inference algorithm outperformed the other methods in most of the environments considered, achieving more than 90% of the optimal performance (the optimal performance was defined as the performance possible in an unrealistic hypothetical situation in which the adapting algorithm 'knows' the level of

the student in advance). In addition to simulating students, simulation was used to evaluate the performance of the ITS (with the various adaptation methods) over the long-term, in situations in which the level of the student is expected to improve over time. We found that in cases where the mean level of the student improves in each step (by a rate of at least 1% after each answer), the item response theory-based method outperforms all of the other methods. In addition, as the improvement rate reaches 1.2% or more, the Gittins-based method is the second best method, while the Bayesian inference method is in third place.

The simulation approach also has limitations: First, when using simulated students the simulation can only check the ability of the adaptive algorithm to correctly assess the students' current level and optimize the predefined evaluation function of the system, but it cannot reflect the students' satisfaction and motivation, nor their learning gain. In addition, an accurate behaviour when given the simulated students' behaviour, does not promise that the accuracy will hold when using real students with different behaviours, which may be less predictable compared to the simulated students. Finally, the simulation does not consider several real-world situations that may transpire with human students, such as fatigue or burnout.

It is also interesting to check the effect of question adaptivity on students' achievements and satisfaction, and evaluate which criteria are more crucial in adapting to the learner's profile. Another interesting question is how can intelligent tutoring components be successfully integrated into widely used MOOC platforms [61,62]. The important questions above should be empirically studied, by determining the challenges and the difficulties in using intelligent educational systems and their effects on the learners' satisfaction rates. In addition, this also can be accomplished by suggesting more personalized learning environments and by developing a more personalized user interface [63].

Recall from Section 2 that a wide range of behaviour of educational systems can be adapted to the learners' profiles. Clearly, the issue addressed in this chapter is the issue of adapting the tasks' hardness to the students' abilities, which is the core of an adaptive system, in order to solve the trade-off between the students' sense of success, which influences their motivation, w.r.t. the need to challenge the students and to train their ability – which can better be done when higher level tasks are presented to them.

Our results can be of interest both to educators and to computer science experts. Educators can learn about the importance of integrating adaptive units in the next generation of educational software, and they can also be aware of the existence of different adaptive schemes, which are appropriate for different situations. Furthermore, computer science experts can be motivated to continue to study the usefulness and effectiveness of different adaptive schemes, or even to study the ability to combine more than one scheme, in order to optimize the system adaptivity for different educational domains and goals. Our main finding is that the Bayesian based method could model a more accurate belief about the students' abilities, while the item response theory based method can better adapt itself in situations where the students' abilities dynamically change during training. Combining these methods with deep learning can assist in a better prediction of the system's ability, as well as a better understanding about task hardness, and a better ability to consider complex student profiles.

7. Conclusions

In this chapter, we describe different adaptivity challenges and concentrate on the challenge of task adaptivity to the students' abilities. We examined different algorithms via simulations i.e., a variation of the classical reinforcement learning method, Bayesian inference method, a Gittins-based method, and a method based on the item response theory. In addition we compared the algorithms' applicability as well as their performance. By simulation we found that in most of the cases, the Bayesian inference-based algorithm achieved the best results. Nonetheless, for environments where the student's level dynamically improves over time, the Rasch+Elo outperformed all other methods. The advantages of the Rasch+Elo method is its easy implementation and sensitivity to changes in the students' abilities, and the fact that it can be adapted to complex environments and to complex student models. Our study can assist designers of ITSs in selecting the optimal adaptation method, given the needs and goals of the ITS, the tutoring domain, expectations regarding students’ skill improvement and the pace of that improvement, etc. Further research should be conducted to verify which evaluation function optimally represents the effectiveness of an ITS, taking into account the preferences of the teachers and students, in order to provide the best individualized tutoring experience possible. In addition, more complicated evaluation functions can be considered and tested, based on the preferences of real students and teachers.

Appendix A

In each step, the algorithm considers all of the possible combinations of the student's parameters, and for each candidate question's level, the algorithm calculates the expected utility, which is calculated as the expected value of the evaluation function when receiving a question at this level, given all of the possible distributions of students. Then, the algorithm chooses the level with the highest expected utility.

$$ChosenLevel = argmax_{level=1..5} \Sigma_{\mu,\sigma} prob(_{\mu,\sigma})* \quad (5)$$

$$(pwins(level \mid \mu,\sigma)*util\ (level)+(1-pwins(level \mid \mu,\sigma))*UtilFailure)$$

where pwins(*level* | μ,σ) is the probability of a question n from level *level* to be answered correctly, util (level) is the utility from a successful answer to a question from this level, and UtilFailure is the negative utility (penalty) resulting from the failure to answer a question from this level. Once a question is chosen and the student's response is observed, the probability of each distribution of the student is updated using the Bayesian rule.

The explanation of the formula is as follows.

1. For each possible *level*, we review the entire table and calculate *pwins(level* | μ,σ*)*, which is the probability that a student with mean level μ and a standard deviation σ will be able to correctly answer a question of the current *level*.
2. Given *pwins(level* | μ,σ*)*, we calculate the expected utility from a question from *level*, if the student's distribution is normal with (μ,σ), where *pwins* is multiplied by the success utility, and *(1- pwins)* is multiplied by the failure utility.
3. Once this value is calculated for each possible level, the level that achieves the highest expected utility is chosen.

Figure A.1 illustrates the Bayesian learning process when used in the ITS. Additional details on the Bayesian inference algorithm are provided in [39].

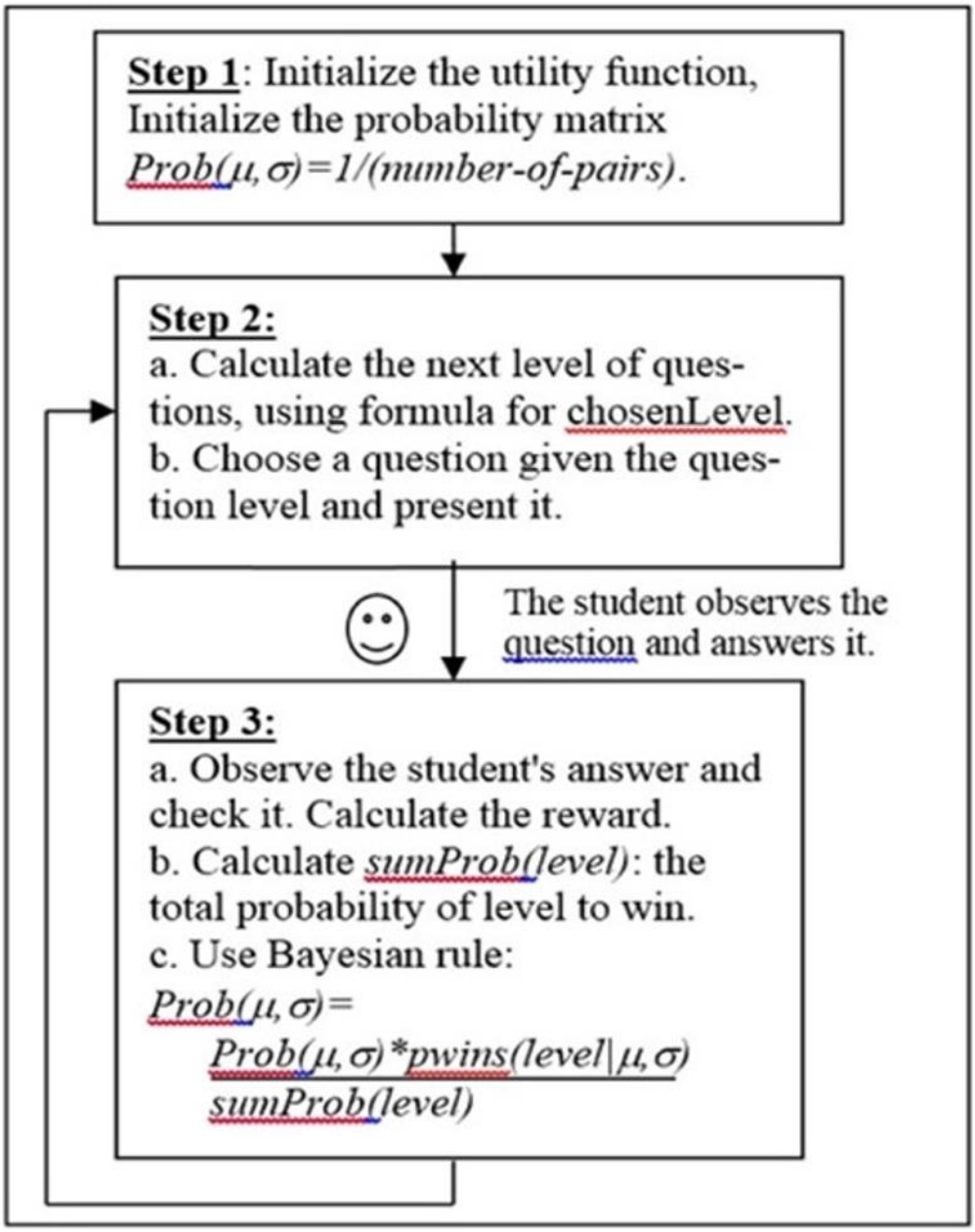

Figure A.1: *The Bayesian learning process.*

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